INDIVIDUAL WEEKLY PROJECT REPORTS

Student: Ashvitha I

Role: Data Collection and Data Quality
Project Title: Flight Delay Analysis and Visualization

WEEK 4 – Identification of Invalid Records and Data Inconsistencies

During the fourth week, I continued examining the dataset to identify unusual values and possible inconsistencies. My focus was on understanding whether the information recorded in the dataset was logically valid and suitable for analysis.

I examined numerical fields related to flight timings and delays and considered whether the values were within reasonable ranges. I understood that certain extreme values could represent genuine flight conditions, while others might indicate errors in the original records.

One important issue discussed during this stage was the presence of negative values in the CRS elapsed time field. Since elapsed time represents the duration between two events, negative values

required additional investigation. I communicated this observation to the preprocessing team so that the records could be reviewed.

I also examined the importance of consistency between related fields. For example, scheduled and actual departure and arrival information should be interpreted carefully when studying delays.

I maintained communication with the team to ensure that the data quality issues identified during this stage were properly understood and documented.

Outcome and Learning:

This week helped me improve my ability to identify inconsistencies in structured datasets. I also learned that data validation requires understanding the meaning of each variable rather than simply checking whether a value is numerical.